

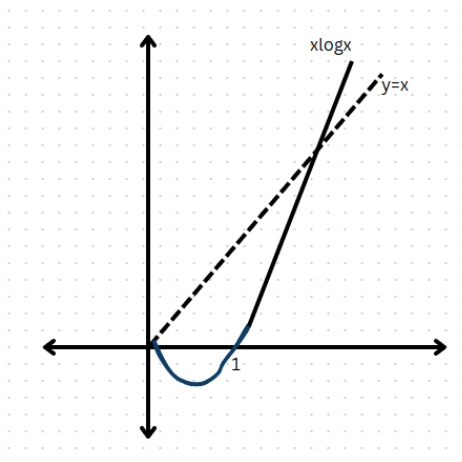
ECE 647: Lecture 9

- Entropy:

$$x \times \log x, x > 0$$

$$f'(x) = \log x + 1$$

$$f''(x) = \frac{1}{x} + 1$$



The source generates symbols according to a distribution P .

$$P\{\text{Symbol } x_i\} = p_i, i = 1, 2, \dots, N$$

Entropy $H = \sum_{i=1}^N P_i \log P_i$ measures the "uncertainty" of the source (ie the amount of information generated by the source).

KL Divergence (Kullback-Leibler):

- Two distributions: P_i and Q_i

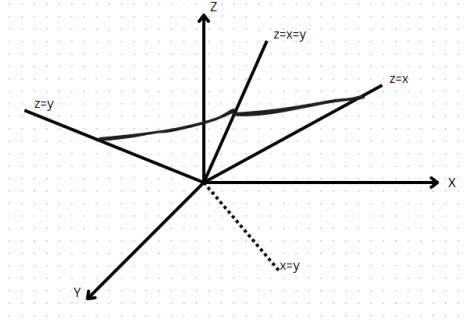
$$D_{KL}(P||Q) = \sum P_i \log \frac{p_i}{q_i}$$
- If q_i is fixed, then this is convex in p_i
- But in fact this is convex in both p_i and q_i (will need the "perspective" of a function later)

- Useful for measuring the "distance" between distributions (See Logistic Regression later)

- Max:

$\max x_i$ is convex

- It is also non-differentiable - Needs the "Max of Convex Functions"



- Log-Sum-Exp:

$\log(e^{x_1} + \dots + e^{x_N})$ is convex.

Proof: Restrict to a line and we use second-order conditions.

$$g(\lambda) = f(\vec{x} + \lambda \vec{y}) = \log(e^{x_1 + \lambda y_1} + \dots + e^{x_N + \lambda y_N})$$

$$\frac{df}{d\lambda} = \frac{y_1 e^{x_1 + \lambda y_1} + \dots + y_N e^{x_N + \lambda y_N}}{e^{x_1 + \lambda y_1} + \dots + e^{x_N + \lambda y_N}}$$

$$\frac{d^2 f}{d\lambda^2} = \frac{1}{(e^{x_1 + \lambda y_1} + \dots + e^{x_N + \lambda y_N})^2} \times \{(y_1^2 e^{x_1 + \lambda y_1} + \dots + y_N^2 e^{x_N + \lambda y_N})(e^{x_1 + \lambda y_1} + \dots + e^{x_N + \lambda y_N}) - (y_1 e^{x_1 + \lambda y_1} + \dots + y_N e^{x_N + \lambda y_N})^2\}$$

Use Cauchy-Schwartz Inequality: $(\sum a_n^2)(\sum b_n^2) \geq (\sum a_n b_n)^2$

Use $a_n^2 = y_n^2 e^{x_n + \lambda y_n}$, $b_n^2 = e^{x_n + \lambda y_n}$, $n = 1, 2 \dots N$

Example:

- High-SNR Approximation of Shannon Capacity:

$$C_i = W \log(1 + \frac{P_i}{\sum_{j \neq i} P_j + N}), \text{ where } P_j \text{ denotes the interference}$$

and N gives the Noise.

At high SNR, $\log(1 + x) \approx \log(x)$

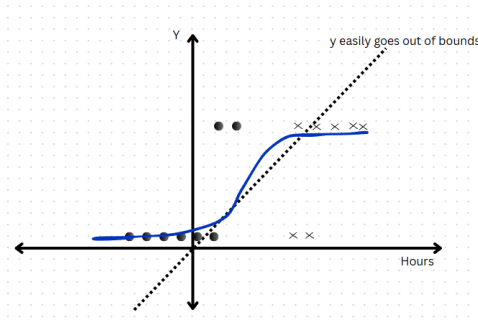
$$C_i \approx W \log P_i - \log(\sum_{j \neq i} P_j + N)$$

This is neither concave nor convex. Let $P - I = e^{x_i}$ then,

$C_i = W \log x_i - \log(\sum_{j \neq i} e^{x_j} + N)$ is concave in $\vec{x}$.

– Logistic Regression:

- * Suppose you have training data (x_i, y_i) where x_i is the explanatory variable and y_i is the categorical variable.
- * We want to learn how to predict y from x . We cannot use Linear regression as it won't be a good fit (shown in the figure), thus instead we use the curve given by: $y = \frac{e^{ax+b}}{1+e^{ax+b}}$



- * This can be thought of as the conditional distribution of y given x .
- * Again, now we need to use a distance to measure how good the match is. We cannot use L_2 distance as $(y - \frac{e^{ax+b}}{1+e^{ax+b}})^2$ is not a convex function. Thus, we use KL Divergence:

- Here, P : Empirical Observation of the Conditional Distribution of y given x , Q : Our estimate above.

$$- D_{KL}(P||Q) = \sum P_k \log \frac{P_k}{Q_k} \text{ is convex in } (P, Q) = \sum_k P_k \log P_k - \sum_k P_k \log Q_k$$

- * We then maximize $\sum_k P_k \log Q_k$
- k denotes a possible outcome. - Suppose the sample is

(x_i, y_i) . Given x_i , our P or Q places probability on two outcomes 1 or 0.

- For P it is: $(1, 0)$ if $y_i = 1$ or $(0, 1)$ if $y_i = 0$. Thus we have: $(1_{y_i=1}, 1_{y_i=0})$

- For Q, it is: $(\frac{e^{ax_i+b}}{1+e^{ax_i+b}}, \frac{1}{1+e^{ax_i+b}})$

- The KL Divergence is then: $1_{y_i=1} \log \frac{e^{ax_i+b}}{1+e^{ax_i+b}} + 1_{y_i=0} \log \frac{1}{1+e^{ax_i+b}}$

* This is the Logistic Regression:

$$\max \sum_i \{1_{y_i=1} \log \frac{e^{ax_i+b}}{1+e^{ax_i+b}} + 1_{y_i=0} \log \frac{1}{1+e^{ax_i+b}}\}$$

* We can verify that it is concave in a.

- Geometric Mean:

$(\prod_{i=1}^N x_i)^{1/N}$ is concave. Proof: See Boyd, P74.

- Quadratic-over-linear:

$\frac{x^2}{y}$ is convex.

Proof: $\delta^2 = \frac{2}{y^3} \begin{bmatrix} y^2 & -xy \\ -xy & x^2 \end{bmatrix}$

Q: Why is it positive-semidefinite?

A: Can also use "perspective-mapping" (later)

- Earlier we have discussed two operations that preserve convexity:

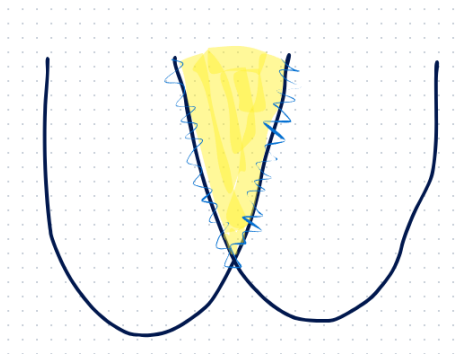
- Non-negative weighted sums - Affine mapping of the arguments

• Point-wise maximum and supremum:

If f_1, f_2 are convex, define: $f(x) \triangleq \max f_1(x), f_2(x)$

Then $f(x)$ is convex.

Proof: The epigraph of $f(x)$ is the intersection of the epigraphs of f_1 and f_2

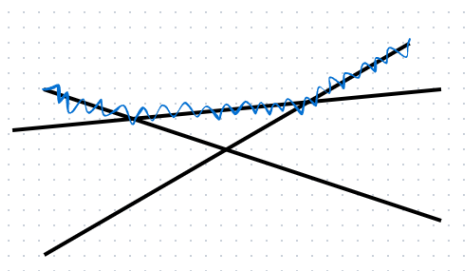


The supremum can be taken over an infinite set. If $f(x, y)$ is convex in x for each y , then: $g(x) = \sup_{(y \in A)} f(x, y)$ is convex in x .

Similarly, point-wise minimum and infimum of concave functions are concave functions.

Examples:

- $\max_i(X_i)$ is convex.
- $\max_i(a_i^T x_i + b_i)$ is convex.
- $\min_i(X_i)$ is concave.
- $\min_i(a_i^T x_i + b_i)$ is concave.



- Sum of the largest r components: For $x \in R^n$, let $x_{[1]} \geq x_{[2]} \geq \dots x_{[n]}$ where $[1], \dots [n]$ is a permutation of 1 to n . Let $f(x) = \sum_{i=1}^r x_{[i]}$. Then $f(x)$ is convex.
 $f(x) = \max(x_{i_1} + \dots + x_{i_r} | 1 \leq i_1 \leq \dots n)$
 - Distance to the farthest point: Let $x \in R^n, C \subseteq R^n$. Let $\|x - y\|$ denote the distance between x and y . Let $f(x) = \sup_{(y \in C)} \|x - y\|$. Then $f(x)$ is convex.
- Q: Infimum of convex functions?